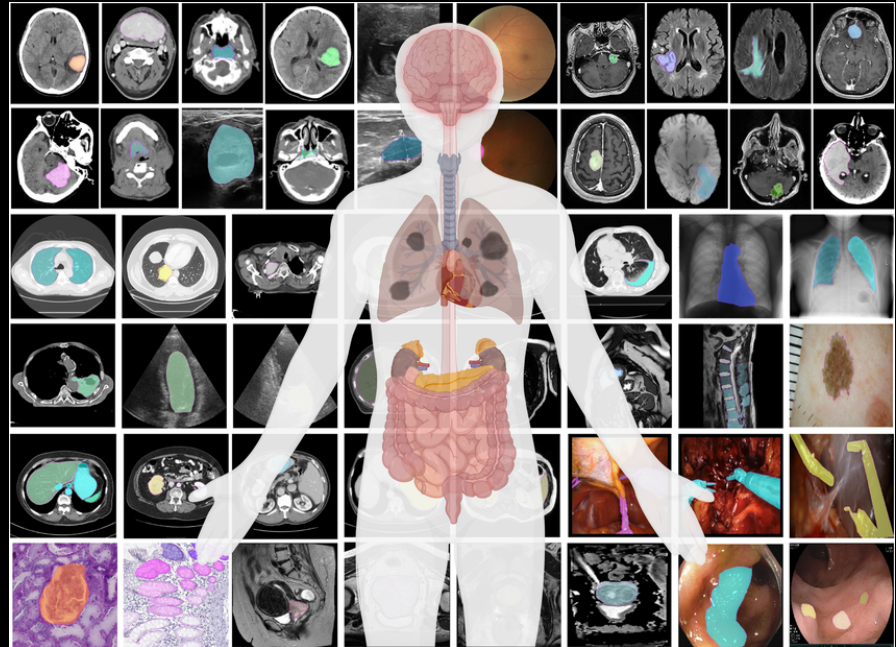
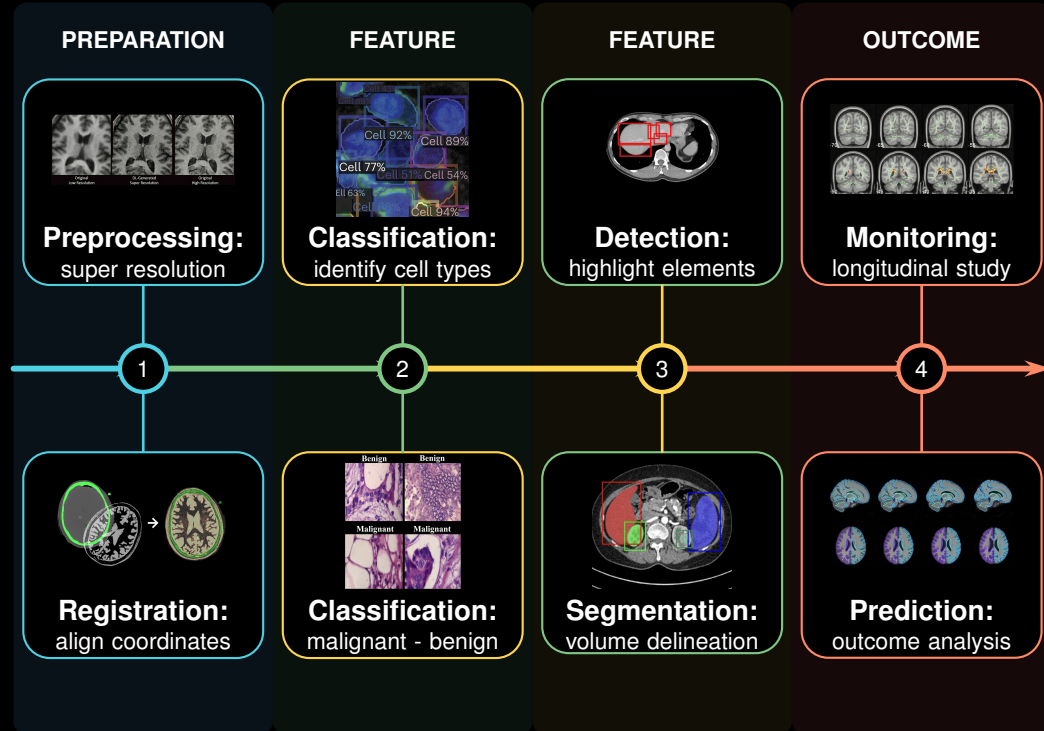


COMMON TASKS IN MEDICAL IMAGE ANALYSIS

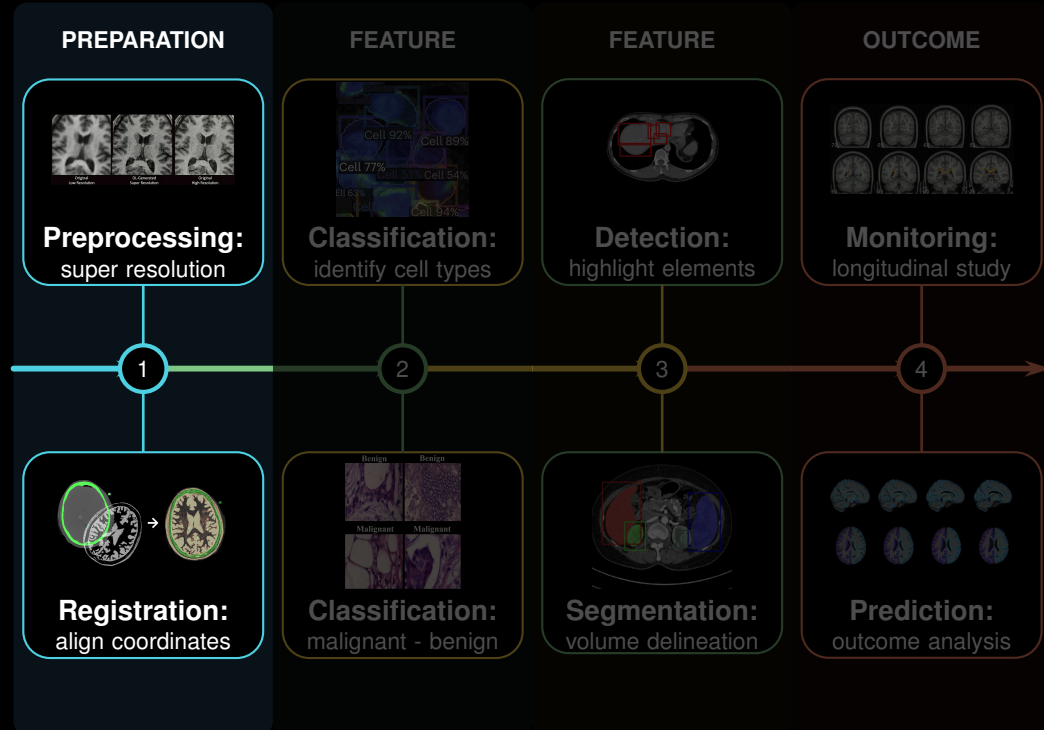


Source: Wang et al., Nature Communications 2024

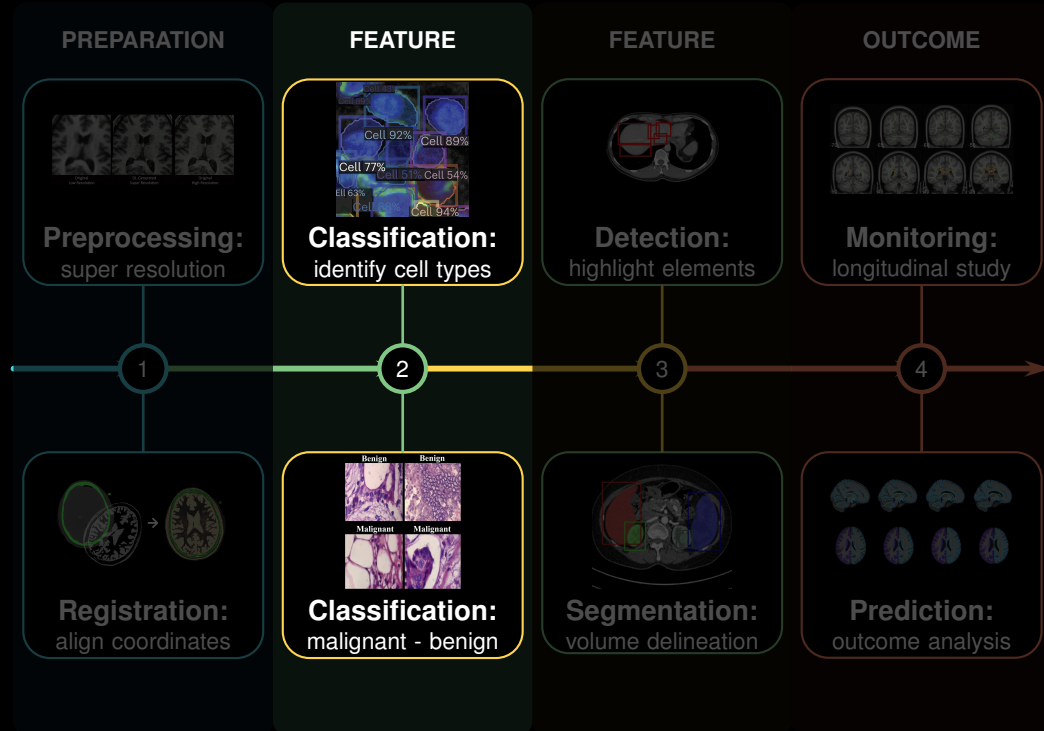
COMMON TASKS IN MEDICAL IMAGE ANALYSIS



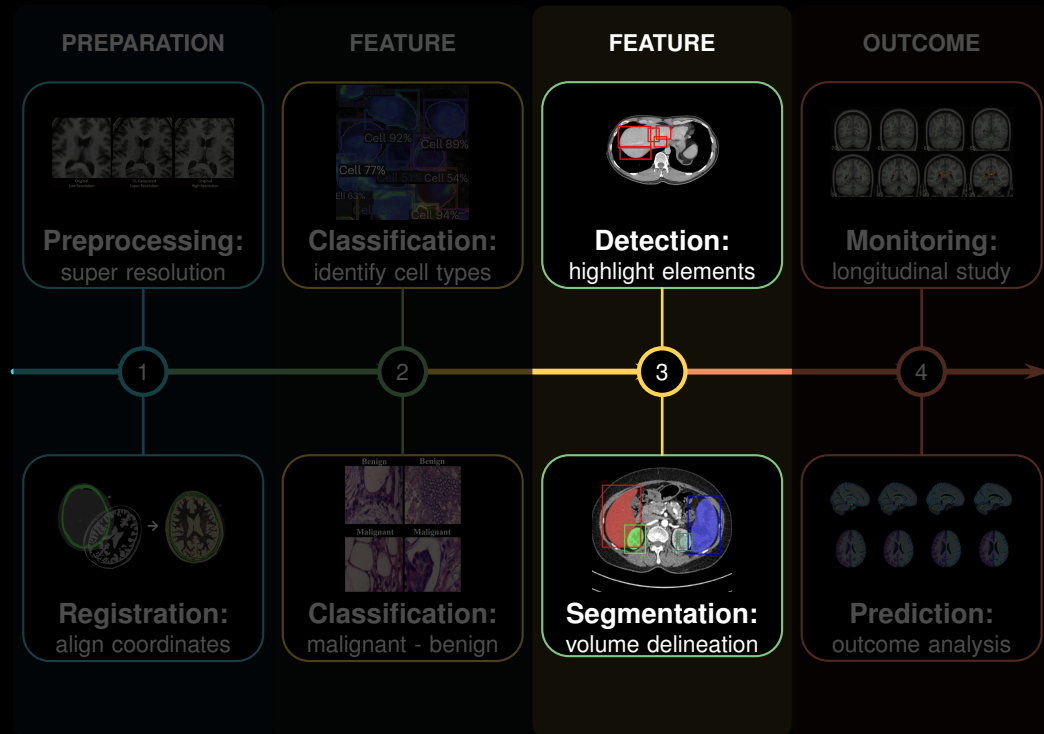
COMMON TASKS IN MEDICAL IMAGE ANALYSIS



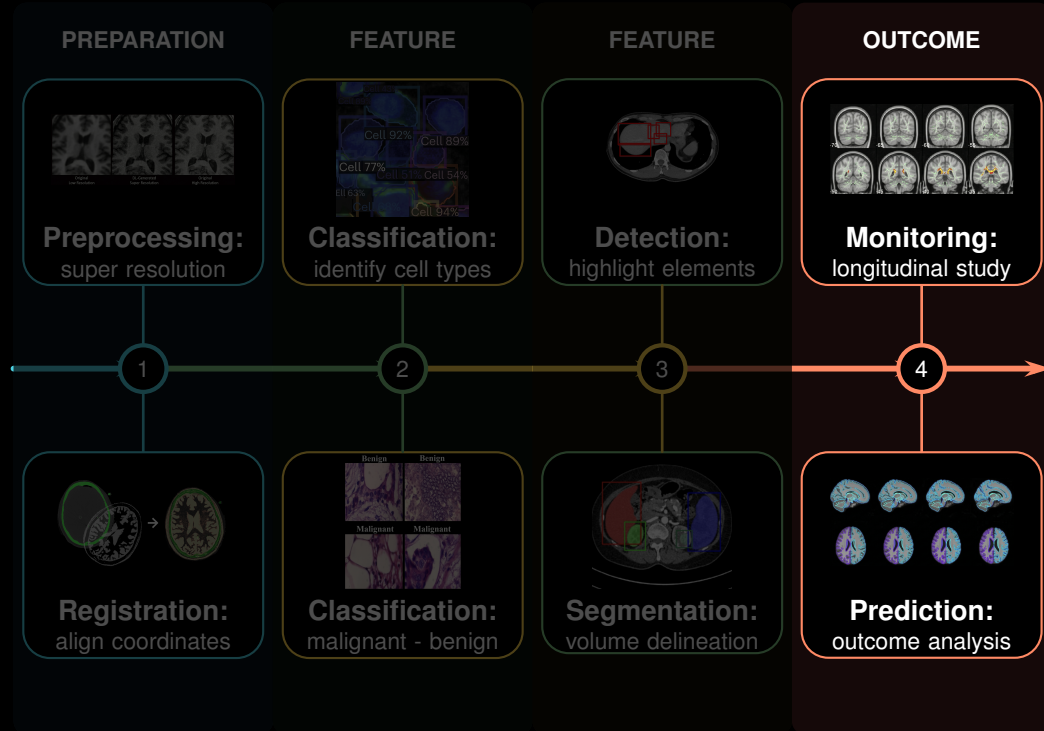
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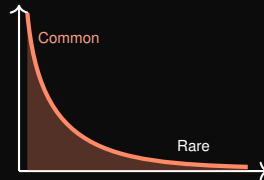


MEDICAL IMAGE DATA IS INTRINSICALLY CHALLENGING

Data Scarcity

Modalities: high-res 3D volumes but few patient samples.

Long-tail: common diseases dominate; rare cases are hard to obtain.



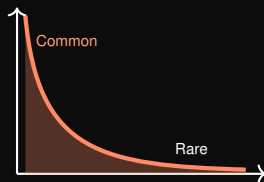
Imbalance: common vs. rare diseases

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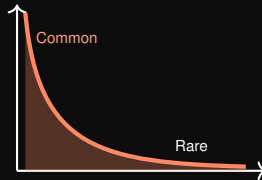
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Imbalance: common vs. rare diseases

Distribution Shift

Non-standard: vendors (Siemens vs. GE), protocols differ.

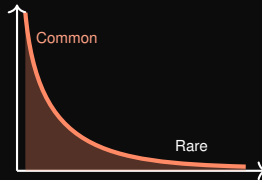
Subjective labels: inter-observer variability.

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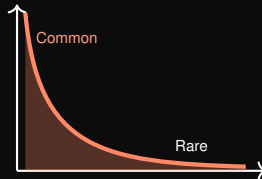
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Privacy Constraints

De-identification risk (e.g., facial reconstruction).

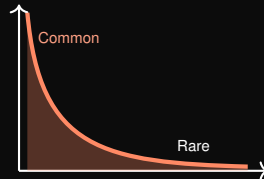
Siloed data due to HIPAA / GDPR constraints.

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